

LEWISTON, ID, USA

WMO: 727830

Lat: 46.375N

Lon: 117.016W

Elev: 438

StdP: 96.18

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24149

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-10.1	-7.0	-15.6	1.0	-8.4	-12.1	1.4	-5.9	12.4	5.7	10.8	5.7	1.8	70	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	37.1	18.5	35.2	18.0	33.2	17.4	19.9	33.3	19.0	32.1	18.1	30.7	3.0	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	11.7	22.4	14.5	10.8	22.0	13.3	10.1	21.6	58.3	33.2	55.1	32.2	52.7	30.7	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	7.9	6.6	-13.3	40.6	3.8	1.7	-16.0	41.8	-18.2	42.8	-20.3	43.8	-23.1	45.0		
(5)			-13.9	21.6	3.8	1.2	-16.6	22.5	-18.8	23.2	-20.9	23.9	-23.6	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	2.3	3.9	7.3	10.7	15.4	19.1	24.6	24.0	18.8	11.3	5.4	1.5		
	DBStd	8.80	4.38	4.03	3.45	3.43	3.99	3.87	3.42	3.39	3.88	3.84	4.09	4.43		
	HDD10.0	978	238	172	96	30	4	0	0	0	1	29	145	263		
	HDD18.3	2791	496	404	343	229	109	35	2	3	41	221	388	521		
	CDD10.0	1735	1	1	12	52	171	273	453	434	264	67	7	1		
	CDD18.3	507	0	0	0	1	18	58	196	178	54	2	0	0		
	CDH23.3	6414	0	0	0	26	248	739	2560	2181	638	22	0	0		
	CDH26.7	3176	0	0	0	4	76	299	1401	1144	249	3	0	0		
(14) Wind	WSAvg	2.4	2.7	2.5	2.6	2.7	2.4	2.5	2.4	2.3	2.1	2.1	2.3	2.6		
(15) Precipitation	PrecAvg	308	30	21	27	31	37	30	16	18	18	24	29	27		
	PrecMax	447	79	56	60	84	96	69	66	75	63	61	68	73		
	PrecMin	191	4	4	6	2	6	5	0	0	0	0	6	3		
	PrecStd	57	19	13	12	18	22	16	15	17	15	14	15	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	14.8	20.4	26.6	31.9	36.6	39.6	39.1	35.3	26.4	17.7	13.8		
		MCWB	8.1	9.1	11.3	14.8	17.4	19.2	19.6	18.5	18.0	15.2	12.2	8.8		
	2%	DB	10.8	12.4	17.2	23.0	28.8	32.7	37.6	36.8	32.1	22.5	14.6	10.5		
		MCWB	6.8	7.4	9.8	13.4	15.9	17.9	18.9	18.1	16.6	13.0	9.8	6.7		
	5%	DB	9.3	10.8	15.2	20.2	26.1	30.1	35.8	34.9	29.5	19.7	12.7	8.7		
		MCWB	5.8	6.6	9.0	12.0	15.0	16.9	18.2	17.6	15.6	11.8	8.8	5.5		
	10%	DB	8.0	9.4	13.2	17.4	23.5	27.6	33.8	32.8	27.1	17.4	11.0	7.1		
		MCWB	4.9	5.7	7.9	10.3	13.9	15.8	17.7	17.0	14.9	10.7	7.5	4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	9.9	12.2	16.0	18.5	20.4	21.8	20.1	19.3	15.7	13.1	9.5		
		MCDB	12.2	13.9	18.4	24.7	28.6	33.8	34.7	35.3	31.4	24.2	17.3	13.5		
	2%	WB	7.2	8.3	10.7	14.0	17.0	18.9	20.2	19.1	17.7	14.0	10.7	7.3		
		MCDB	10.4	11.4	15.6	21.7	26.4	30.8	33.5	33.5	28.6	20.4	13.7	10.1		
	5%	WB	6.2	7.0	9.7	12.4	15.8	17.8	19.3	18.4	16.6	12.7	9.3	5.8		
		MCDB	8.9	10.2	14.5	19.2	24.0	28.3	32.4	32.2	27.5	18.5	12.0	8.1		
	10%	WB	5.1	6.0	8.4	10.9	14.7	16.6	18.4	17.7	15.4	11.5	8.0	4.8		
		MCDB	7.6	8.9	12.8	16.5	22.1	25.8	31.6	30.7	25.8	16.4	10.6	6.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	7.6	9.7	10.9	12.1	12.8	15.7	15.5	14.0	10.5	7.0	5.5		
		MCDBR	7.3	9.2	12.8	15.8	16.5	17.4	18.6	18.5	17.9	13.9	8.8	7.0		
	5% WB	MCWBR	4.7	5.6	7.3	8.1	7.0	6.6	6.6	6.8	7.2	6.9	5.4	4.6		
		MCWBR	6.9	8.6	11.6	14.4	14.6	15.7	16.6	16.5	16.0	12.2	7.9	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.299	0.318	0.340	0.358	0.358	0.359	0.367	0.347	0.329	0.300	0.293		
		taud	2.471	2.505	2.488	2.420	2.393	2.412	2.413	2.384	2.418	2.472	2.509	2.458		
	Ebn at Noon		811	886	914	918	908	906	900	878	867	833	798	768		
		Edh at Noon	67	78	92	108	115	113	112	111	98	80	64	61		
(44) All-Sky Solar Radiation	RadAvg		1.25	2.14	3.25	4.75	5.93	6.71	7.52	6.21	4.54	2.79	1.45	0.98		
	RadStd		0.11	0.24	0.32	0.41	0.47	0.48	0.34	0.40	0.36	0.25	0.12	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+97	+60
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+87	N/A

Nomenclature: See separate page